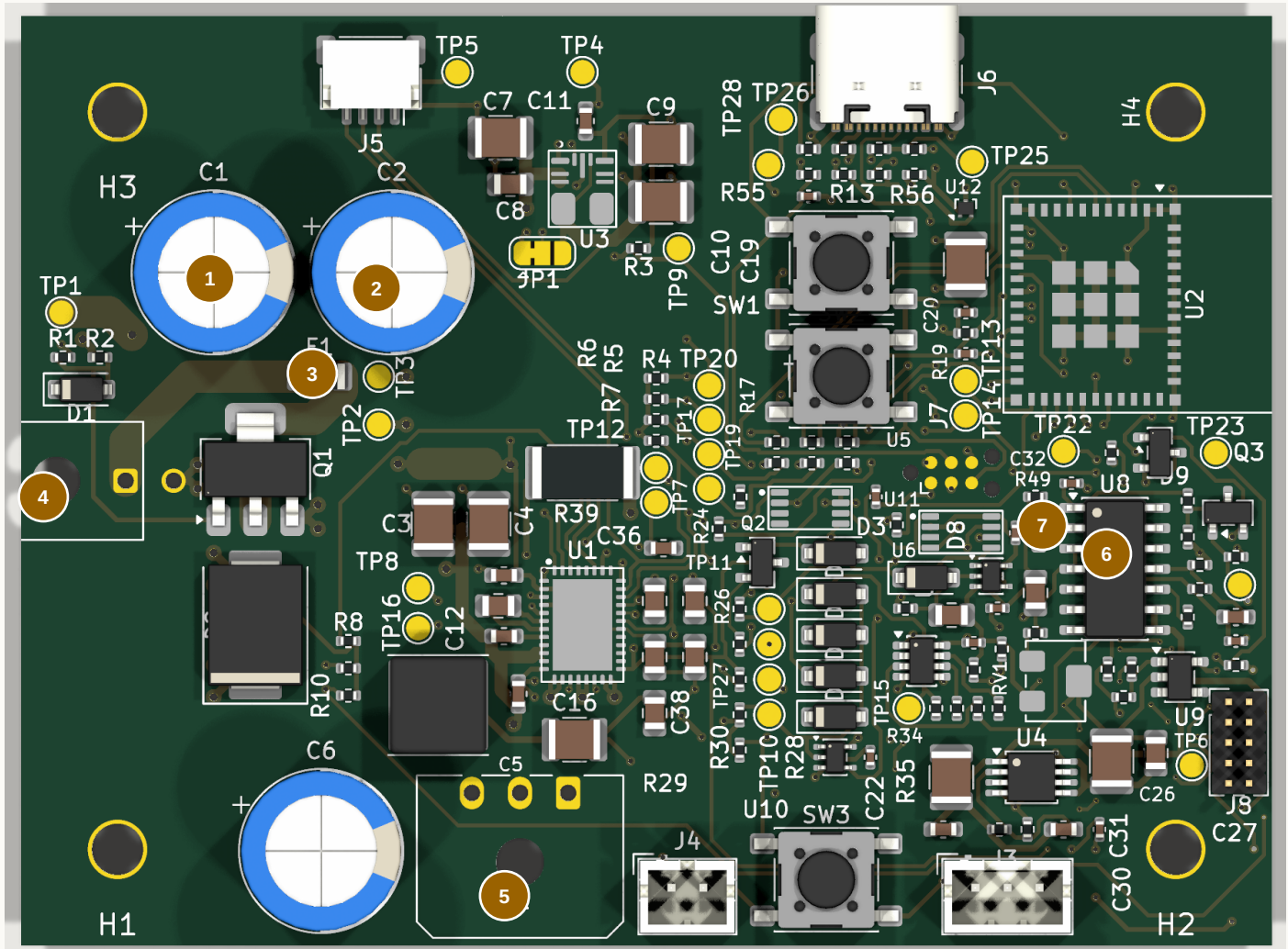


1A PCB-01 Population Map

FIELD GUIDE

TOP / COMPONENT SIDE - KiCad target appearance



1 C1
5 J2

2 C2
6 U8

3 F1
7 C34

4 J1

SOLDER ORDER: U8 -> C34 -> J1/J2 -> C1/C2 -> INSPECT -> F1 LAST

1 + 2 C1 / C2

EEU-FR1H471, 470 uF 50 V. Positive lead to board +; negative stripe to pin 2 / PGND. Seat square without crushing bung.

3 F1 - INSTALL LAST

Low-profile tinned copper link across the 1206 pads. Near-zero ohms after soldering. External wall-side 3 A fuse remains mandatory.

4 + 5 CONNECTORS

J1 43045-0200; J2 43650-0300. Larger chisel tip. Tack one electrical pin, square and fully seat housing, then finish. Locating pegs are not soldered.

6 U8 / 7 C34

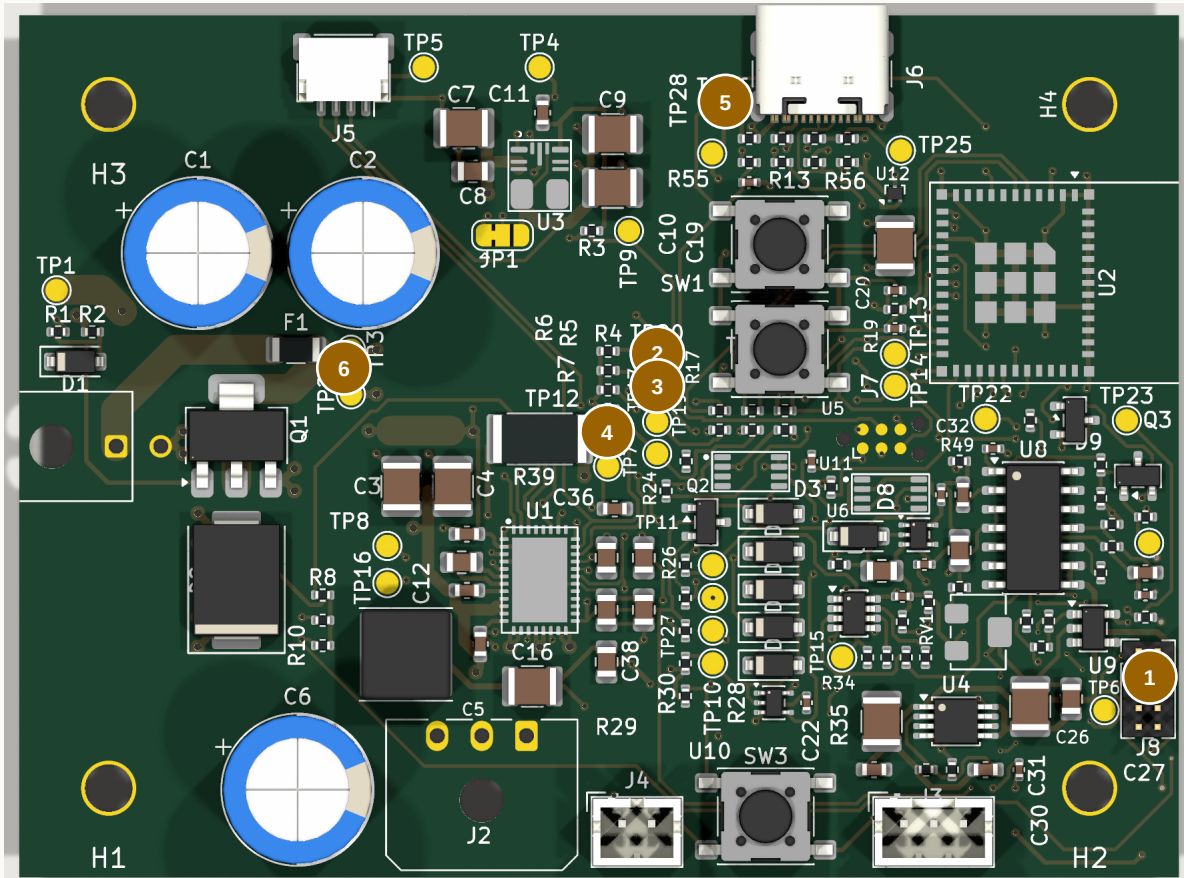
U8 LM2907M/NOPB: notch/dimple to pin-1 marker, tack pin 1 then diagonal pin 8, flux and drag-solder. C34 C1206C104K3GACTU: 100 nF COG 25 V +/-10%, spanning the smaller 0603 site, not shorting it.

PCB-01: USE THE IRON, NOT BROAD HOT AIR. Then clean and dry; inspect under the microscope; verify no hard short from RAW24, VM24, 3V3, or 12V_TACH to ground; photograph the board.

1B Removable Tuning Leads

FIELD GUIDE

INSTALL ON THE BENCH BEFORE THE BOARD GOES OVERHEAD



1 J8 SOX/SPEED
4 TP12 DRVOFF

2 TP20 FG
5 TP26 AGND

3 TP17 NFAULT
6 TP2/TP3 VM24

NORMAL TUNING: LOW-VOLTAGE PIGTAILS

Use 30 AWG insulated wire, 50-100 mm. Fit J8.9 SOX (white), J8.6 SPEED (green), TP20 FG (yellow), TP17 NFAULT (orange), TP12 DRVOFF (blue), and TP26 AGND (black). Label both ends. Make insulated test loops at the free ends. For the OWON, CH1 is SOX and CH2 selects FG/SPEED/NFAULT/DRVOFF; both channel grounds use AGND.

VM24: DIRECT x10 PROBE

Do not add a long VM24 pigtail. With power off, attach the x10 tip directly to TP2 and its spring ground to TP3. Strain-relieve the probe before mounting. In a bus session, every OWON ground must use PGND.

FIT, INSPECT, REMOVE AFTER FINAL TUNING

All sources removed and bulk capacitors discharged. Solder once, route insulated wire flat, add Kapton strain relief away from each joint, inspect under magnification, then continuity-check every lead to its named net and for adjacent-pad shorts. Cap every free end before power. TP4 is also usable AGND; never wire motor phases, switch nodes, or USB data pads. Remove the pigtails after the complete exposed and housed tuning campaign, then clean and inspect.

OWON CHANNEL GROUNDS ARE COMMON. Never put one ground on AGND and the other on PGND in the same setup.